

A hand-motion dataset can test compositional generalisation only when its labels describe the scored motion

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Abstract

Held-out label combinations are used to test whether motion models generalise to unseen compositions, such as a known action performed with a known tool in a combination absent from training. On observational data the split is made on labels, and whether holding out a label also holds out the motion has not, to our knowledge, been examined. We measured a paired compositional penalty, the reduction in reconstruction error gained by a prior that had seen the held-out combinations, on four public hand-motion datasets. It was calibrated with a synthetic dataset whose held-out pairs are interpolable by construction, planted interactions of known size, and permuted label grids on three of the real datasets. The synthetic control returned +2.6 % of the naive error, and the permuted grids returned +3.7 % on OakInk-Image, +1.4 % on GRAB and -3.4 % on TACO, so the floor of the measurement depends on the dataset. On OakInk-Image, joining clips so that each correct label described only the first half of a trajectory, with the number of trajectories, the categories and the training volume held fixed, reduced the penalty from +17.0 % to +3.9 %, not distinguishable from the control (difference +0.9, 95 % CI -1.1 to 2.9). The windows that the label did describe lost 82 % of their penalty above the control, so the change lay in what the training sets contained rather than in what was scored. A penalty on TACO, predicted before its data were opened, was confirmed (+10.6 %), and per-seed estimates of it ranged from -3.8 % to +27.4 %. A hand-motion dataset can therefore test compositional generalisation only when its labels describe the scored motion, and the result should be read against within-dataset references and over many splits.

1. Introduction

Motion priors for dexterous hands are trained on recorded human motion and are then asked to produce motion that the recording did not contain. Priors with a continuous latent space [1] and priors built from a bank of primitives both rest on the expectation that what was learned separately can be recombined, so that an action recorded with one tool and a tool recorded in another action remain usable together. The usual test holds out combinations of labels: every action and every tool occurs in training, but some combinations do not, and the model is scored on those combinations. The design is well established in language [2, 3], in vision [4, 5], in robot manipulation [6, 7, 8] and in hand-object interaction, where TACO holds out the tool-action-object triplet while keeping every tool in training [9] and verb-noun splits have been built on H2O and FPHA [10]. This study adopts that design and does not propose it.

A split of this kind is made on labels, and it holds out the motion only if each label describes the motion it is attached to. In a designed or generated benchmark this is true by construction. Public hand-motion datasets are observational: their annotations were made for other purposes and at a granularity chosen for those purposes. An OakInk-Image trajectory is a single grasp lasting a median of 2.4 s. A TACO trajectory is described by its authors as one tool-use action and lasts a median of 4.9 s. A GRAB trajectory is a whole interaction of 8.5 s including approach and release, and an OakInk2 trajectory is a complete task of 82 s that contains several primitives. A prior is trained and scored on windows of about

one second, so on the longer recordings a label can be correct and still describe only part of what is scored. Motion that belongs to a held-out combination can then enter training under another label, and much of what is scored as the held-out combination can be something else.

A second difficulty is that the measurement has no known zero. A compositional penalty is a difference between two trained models, and two models trained on different data differ even when the held-out combinations are predictable from what each has seen. Held-out-triplet results for hand-object motion forecasting have been reported from one partition and one run per split [9], so neither the size of this floor nor the spread over splits is known.

The purpose of this study was to determine when a held-out label combination measures compositional generalisation on hand-motion data. One paired measurement was calibrated with three kinds of control and applied to ten axes of four public datasets. The relation between label and scored motion was then changed on one dataset with everything else held fixed, and a penalty was predicted on a fourth dataset before its data were opened. Three questions were addressed: (1) how large the paired measurement is when the true penalty is zero, on synthetic and on real data, and whether it responds to a penalty of known size; (2) whether a penalty survives when a correct label no longer describes the whole scored trajectory, and whether it is lost in the training sets or in the scoring; and (3) whether a penalty predicted on a dataset that played no part in forming the question is found.

The contributions are as follows. First, a calibrated protocol for held-out-combination evaluation on observational motion data, combining a zero-truth control, planted positive controls and permuted-grid controls, with the finding that the floor they establish differs between datasets; each control has precedents outside compositional evaluation, named in Section 2. Second, a controlled demonstration that correct labels which describe only part of a trajectory remove a compositional penalty, with the loss located in the training sets rather than in the scored windows. Third, a confirmed out-of-sample prediction on TACO together with the per-seed spread of the estimate. Fourth, every sweep run under the protocol is listed with its result (Appendix D), and the per-seed results, gate transcripts and written readings of every sweep are supplied as supplementary material.

2. Related work

Held-out combinations. Compositional generalisation is commonly measured by withholding combinations whose parts were seen. SCAN [2] and CFQ [3] do so for language, the latter varying how strongly the test distribution diverges from training across several splits; Hupkes et al. [11] review the practice. In vision, Something-Else holds out verb-object combinations in video [4], and in robot learning unseen combinations of environmental factors [6], of atomic skills [7] and of instructions and contexts [8] have been used to evaluate policies. In hand-object interaction, TACO evaluates motion forecasting on held-out triplets [9] and Tse et al. [10] build compositional verb-noun splits on H2O and FPHA for action recognition.

When holding out a label does not hold out the target. Finegan-Dollak et al. [12] showed that splitting text-to-SQL data by question leaves the SQL queries being predicted in both training and test, and proposed splitting by query instead. SCAN contains a related case: models trained with "turn left" seen only as a bare command still generalised to its compositions, which the authors attribute to its output action appearing in training inside other commands, whereas they failed on "jump", whose action appeared nowhere else [2]. In zero-shot action recognition, classes treated as unseen can reach pre-

training under other names [13, 14, 15], and in zero-shot detection training images that contain unseen objects are removed [16]. In zero-shot temporal activity detection, training clips that contain an unseen activity are removed as a matter of protocol [17]. Apart from that protocol, these works define overlap by class identity or by name, and a survey of leakage in machine-learning-based science does not list partial label coverage among its sources [18]. Label coverage has also been examined as a property of motion data: a single sequence label leaves most of a motion-capture sequence undescribed [19], and the temporal bounds of hand-object interactions change recognition accuracy [20]. Label quality has been audited on an observational compositional benchmark [5], and benchmarks built from the same data have been shown to disagree [21]. What this study adds is the case in which every label is correct but describes only part of a scored trajectory, shown by a controlled manipulation on motion data and located in the training sets.

Calibrating a measurement. Each control has an origin outside compositional evaluation. A condition whose true effect is zero is the negative control of epidemiology [22] and the basis of empirical calibration of estimates [23, 24]; in machine learning, the bias of a replicated test set has been measured in the same spirit [25]. Permutation nulls that preserve structure are standard for classifier evaluation [26] and have been used for generalisation across conditions in neuroscience [27]. Planting an effect of known size and measuring its recovery is the injection-recovery test of astronomy [28], the spike-in of genomics [29] and, in machine learning, the planted artefact [30]. To our knowledge they have not been combined to calibrate a held-out-combination gap.

3. Methods

3.1 Datasets and hand representation

Four public datasets were used: OakInk-Image [31] (770 trajectories at 30 fps, median 72 frames), GRAB [32] (1,048 trajectories at 30 fps, median 254 frames, from subject archives s1 to s7 and s10), OakInk2 [33] (609 trajectories, subsampled to 7.5 fps, median 613 frames) and TACO [9] (2,317 trajectories at 30 fps, median 148 frames). OakInk-Image and OakInk2 come from the same collection effort, so the four are not independent samples of how a hand-motion dataset can be built. Each provides MANO hand parameters [34]. For the right hand, each joint rotation was converted to Euler angles and projected onto a 27-degree-of-freedom anatomical model (four per finger, five for the thumb and six for the wrist). The flexion axis, its sign and the abduction axis were inferred from each dataset, separately for fingers and thumb, by joint-limit satisfaction, and were not assumed (Appendix A). Poses were normalised to [-1, 1] by the joint-limit box, so that the scale is the same on every dataset.

The labels used were these. On OakInk-Image a trajectory's fine label is its object identity and intent; the object's category, functional class and affordance come from the dataset's own taxonomy [31], and for one axis the capture subject was added to the fine label. On GRAB the fine label is the object and the intent verb of the sequence name; the shape class is a grasp-relevant grouping of the 51 objects into 9 classes, and the intent class is a grouping of the verbs. On OakInk2 the fine label is the recording's sequence of annotated primitives; the scene comes from the recording's metadata and the verb is the leading verb of its task sentence. On TACO the fine label is the tool-action-object triplet.

3.2 The measuring prior

The prior follows LAMP [1]; we adopt its design and do not propose one. A 32-frame window of 27 degrees of freedom was encoded into a 12-dimensional Gaussian latent per frame by two-layer

bidirectional gated recurrent units [35] of 256 units and reconstructed by unidirectional ones under a bounded output (2,283,059 parameters). It was trained as a beta-VAE [36, 37] for 120 epochs on windows cut every 4 frames, with mean-squared error on pose, a first-difference term weighted 0.1 and a free-bits floor of 0.02 nats per dimension. Beta rose linearly to 1.0 over the first 12 epochs, the learning rate was 0.001 and batches held 256 windows. The first 15 % of each arm's training trajectories were used as a validation set, and the checkpoint with the lowest validation loss was scored. A window spans 1.07 s at 30 fps and 4.3 s on OakInk2. The prior is unconditional: it never receives a label, and labels serve only to define the split.

The error throughout is an autoencoding error and not a generation error. Target windows were cut every 16 frames. Each was passed through the encoder, the latent was taken at its mean, and at every step the decoder received that step's latent together with the window's true first frame; no other ground-truth frame was given. The error was the mean-squared error between output and window over all 27 normalised degrees of freedom. The penalty therefore measures how much less well a window of an unseen combination passes through the prior's bottleneck when that combination was absent from training; it does not measure whether sampling from the prior produces the combination.

3.3 The paired compositional penalty

A composition was one cell of a two-factor label grid, formed by grouping fine labels: on OakInk-Image, for example, the object identity was replaced by its category, functional class or affordance. On OakInk2's transitions axis a composition was instead an ordered pair of consecutive primitives within a recording's sequence.

For each seed, four to six cells were held out and about half of the trajectories in those cells, taken as whole fine labels, formed the target set. Two priors were trained on the same number of trajectories (the budget): the naive arm's training data excluded the held-out cells, and the informed arm's included the remaining trajectories of those cells, up to half its budget. Both were scored on the identical target windows, and the penalty was the naive error minus the informed error, expressed as a percentage of the naive error. Both arms share their targets, so differences in target difficulty cancel. The budget was 256 trajectories per arm unless stated. Because it counts trajectories, the training volume per arm differs between datasets (Table 2). Each seed drew its own split and its own random fill of both arms, and the arms shared 7 % to 42 % of their training trajectories.

3.4 Gates

Four checks were applied before a sweep was trusted. No target's exact fine label could appear in either training set (label leak). The informed arm had to reach more than 0.8 of the held-out cells with a mean of at least three trajectories per covered cell (coverage); seeds that failed were kept, and estimates without them are given where they bear on a claim. No target frame could appear verbatim in either arm's training data, checked on five seeds of each derived dataset (frame leak). For trajectories built by joining clips, no clip of a target trajectory could carry a fine label seen by either arm (constituent leak). Every split in Table 2 was re-derived from its stored seed and compared with the stored record, with no mismatch in 1,920 checks.

3.5 Controls

Zero-truth control. A synthetic dataset of 3,000 trajectories at 30 fps (mean 211 frames) was built from nine grasp poses. A trajectory consisted of three to seven segments between poses drawn at random, each

segment following a minimum-jerk profile $w(t) = 10t^3 - 15t^4 + 6t^5$ as the convex combination $(1 - w)a + wb$ of its end poses a and b ; all 81 ordered pairs of poses occurred. Each end pose was perturbed by a draw of scale 7° from a fixed six-dimensional subspace, the wrist followed its own random path, and independent noise of 0.4° was added to every frame before the joint coupling and limits were applied. A segment is therefore determined by its end poses up to that noise, so a pair never seen in training can be reproduced from poses that were, and the true penalty is zero by construction in the data. The held-out unit is an ordered pair of poses rather than a cell of a two-factor grid, which is why permuted grids were added.

Planted controls. A second synthetic dataset added to each segment an offset fixed for its ordered pair and independent of its end poses, drawn at a scale of 18° from the same subspace and applied as $\sin(\pi w)$, so that it can be learned only from trajectories containing that pair. On GRAB, a 27-dimensional offset was drawn for each shape-by-intent cell, its additive part was removed by double centring so that only an interaction remained, and it was scaled to the synthetic offset's per-frame RMS and added to every trajectory of the cell; 53 of 70 cells received a non-zero offset. The GRAB planted sweep ran at a window stride of 32, as did its unmodified comparison.

Permuted-grid controls. On OakInk-Image and GRAB, the second factor was permuted over fine labels, so that the number of cells, the held-out counts and the candidate cells were those of the real grid but a cell no longer grouped trajectories that share their factors. TACO's action-by-tool grid is 16 % occupied, because a tool affords few actions, and permuting one factor left at most 3 usable cells against 17. On TACO the cell labels were therefore shuffled over the 151 triplets, which keeps every cell's size exactly and scrambles both factors.

3.6 Changing whether a label describes the scored motion

On OakInk-Image, two clips were joined into one trajectory labelled by its first clip, with a four-frame linear cross-fade and no clip used more than once. In the aligned version both clips had the same object and intent, so the label described every window. In the misaligned version the second clip had the same object category but another intent, so the label described the first clip only. Both versions had 334 trajectories over the same 33 object categories, with per-category counts within one of each other, mean lengths of 204 and 200 frames, and 2,787 and 2,728 expected training windows per arm. Because 770 clips cannot supply non-overlapping trajectories at a budget of 256, both were swept at a budget of 64, with the unmodified clips and the zero-truth control at that budget.

To locate where the penalty was lost, both versions were re-run on the same seeds with every target window classified as lying wholly in the first clip, wholly in the second clip or across the join, and the penalty was computed within each class. If the penalty fell only because windows of other motion were scored, first-clip windows of the misaligned version would keep the aligned version's penalty; if it fell because the training sets changed, first-clip windows would lose it as well.

3.7 An out-of-sample prediction

Before any TACO file was opened, it was written down that TACO's confirmatory axis would carry a penalty above the zero-truth control. The axis was fixed by a rule written at the same time: the first of action by tool, action by object and tool by object that passed the gates of Section 3.4 with coverage failing on at most 10 of 40 seeds, at the largest number of held cells from five, four and three. The rule

selected action by tool with five held cells; the other two axes were not run. The TACO permuted-grid control was declared before the confirmatory sweep started.

3.8 Statistics and written readings

A seed draws a split, fills both arms and trains both priors, so each seed contributes one penalty. Penalties were compared by two-sided Welch tests [38] on per-seed values, with 95 % confidence intervals on the difference of means, and against the zero-truth control at the same budget rather than against zero. These tests quantify variation over seeds of one fixed dataset; comparisons between datasets are comparisons of fixed objects. Four contrasts were declared primary: TACO against the control, the OakInk-Image and TACO permuted grids against their real grids, and the misaligned against the aligned OakInk-Image version. All four remain below $p = 10^{-5}$ after a Bonferroni correction for four tests; the other tests are descriptive.

The settings and readings of every sweep in Sections 4.1, 4.3 and 4.4, and of the rows of Table 2 marked as such, were written down before training in documents supplied with the results. They are internal documents with file dates, not registrations time-stamped by a third party. The other rows of Table 2 come from earlier sweeps of the same protocol. Seeds were never added to a sweep after its result had been seen, with one exception listed in Appendix C.

4. Results

4.1 Calibration

The zero-truth control returned +2.59 % of the naive error over 20 seeds at a budget of 256 (SD 2.30) and +2.96 % at a budget of 64 (SD 2.34, 20 seeds), both above zero (Table 1). The planted synthetic control returned +9.12 % over 18 seeds, +6.5 points above the zero-truth control (95 % CI 4.5 to 8.6, $p = 4.7 \times 10^{-7}$). The planted interaction on GRAB returned +13.15 % over 40 seeds at a stride of 32, against +0.71 % for the same sweep without it ($p = 3.1 \times 10^{-9}$) and +10.6 points above the zero-truth control ($p = 3.2 \times 10^{-11}$). Only one size of planted interaction was run, so this shows that an interaction of that size is detected on GRAB's poses, not the smallest one that would be.

Table 1. Calibration. Penalty as a percentage of the naive error; budget 256, window stride 4, 40 seeds unless stated. The real grid of OakInk-Image category by intent has 69 seeds.

Condition	Penalty (%)	SD	Seeds positive	Compared with	Difference (95 % CI)	p
Zero-truth synthetic (n = 20)	+2.59	2.30	17/20	zero		
Planted synthetic (n = 18)	+9.12	3.66	18/18	zero-truth	+6.53 (4.47 to 8.58)	4.7×10^{-7}
OakInk-Image category × intent, permuted	+3.65	6.79	27/40	zero-truth	+1.07 (-1.32 to 3.45)	0.37
OakInk-Image category × intent, permuted	+3.65	6.79	27/40	real grid	-11.53 (-14.45 to -8.62)	5.8×10^{-12}
GRAB shape × fine intent, permuted	+1.38	6.76	24/40	zero-truth	-1.21 (-3.58 to 1.17)	0.31
GRAB shape × fine intent, permuted	+1.38	6.76	24/40	real grid	+0.06 (-2.50 to 2.62)	0.96
TACO action × tool, permuted cells	-3.39	6.46	10/40	zero-truth	-5.97 (-8.27 to -3.68)	2.9×10^{-6}
TACO action × tool, permuted cells	-3.39	6.46	10/40	real grid	-14.01 (-17.34 to -10.69)	2.3×10^{-12}

The floor of the measurement therefore depends on the dataset (Table 1, Figure 1). On OakInk-Image the permuted grid did not differ from the synthetic control. On GRAB it equalled the real grid, although its informed arm was the more exposed to held-out cells (0.38 against 0.30). On TACO it read below the synthetic control: there the informed arm, which swapped in unrelated triplets, was worse than the naive arm (mean error 0.0303 against 0.0293). The synthetic control is therefore one reference rather than the zero of the measurement, and on TACO the more demanding reference was used, as declared before the sweep: the larger of the synthetic control and the permuted grid. The positive floor on synthetic data need not be a bias of the design alone, because a recurrent prior need not reproduce minimum-jerk interpolation exactly on pairs it has not seen; the paired design cannot separate the two.

On OakInk-Image and TACO the real grids lay 11.5 and 14.0 points above their permuted grids. A permuted cell is a scattered union of fine labels, whereas a real cell groups trajectories that share their factors and may occupy a compact region of pose space, so these contrasts show that the penalty depends on which fine labels are grouped into a held-out cell, not that it arises from composition as distinct from an unseen region of pose space.

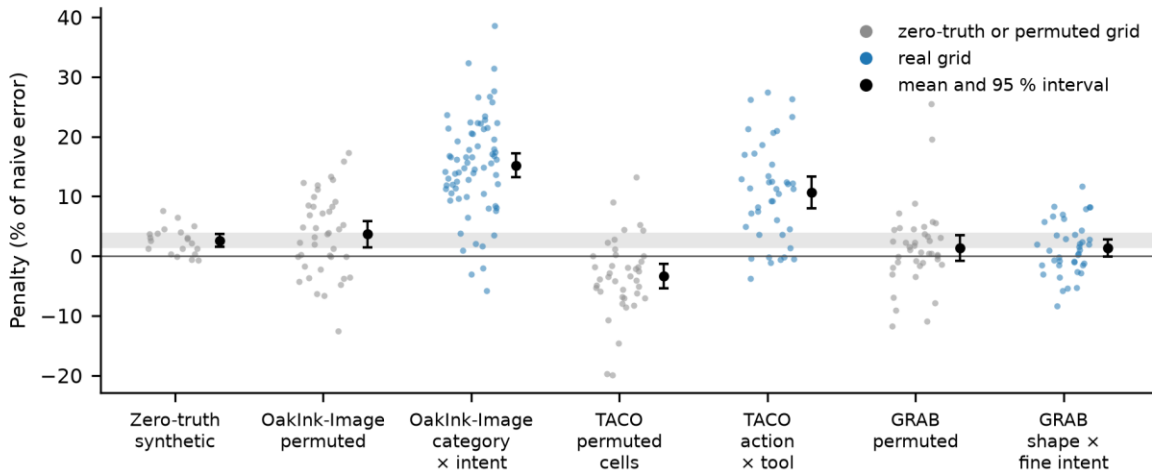


Figure 1. Per-seed penalty for the zero-truth control, three permuted grids and the real grids they permute. Each point is one seed; black markers are means with 95 % intervals; the grey band is the 95 % interval of the zero-truth control's mean. Budget 256, stride 4.

4.2 Penalty by dataset and axis

Table 2. Compositional penalty by dataset and axis at a budget of 256. Windows is the expected number of training windows per arm. Held cells is the number of cells held out per seed. The 95 % CI is of the mean over seeds. Naive error is the naive arm's mean-squared error on the target windows, the denominator of the penalty. Informed exposure is the share of the informed arm's trajectories drawn from held-out cells. p is from Welch's test against the zero-truth control. W marks rows whose settings and readings were written down before training. Rows marked † have 12 seeds; a replication of each on 40 fresh seeds is running and no claim rests on them.

Dataset	Axis	Windows	Held cells	n	Penalty (%)	SD	95 % CI	Seeds positive	Naive error	Informed exposure	p
OakInk-Image	functional class × intent †	4,591	4	12	+17.9	14.0	+9.0 to +26.8	11/12	0.0840	0.32	0.0029
OakInk-Image	category × intent	4,591	5	69	+15.2	8.3	+13.2 to +17.2	66/69	0.0672	0.23	< 10 ⁻¹²
OakInk-Image	affordance × intent	4,591	5	130	+12.2	8.2	+10.8 to +13.6	124/130	0.0682	0.24	< 10 ⁻¹²
OakInk-Image	category × subject W	4,591	4	40	+10.3	9.0	+7.4 to +13.2	37/40	0.0674	0.16	5.9 × 10 ⁻⁶
TACO	action × tool W	8,144	5	40	+10.6	8.3	+8.0 to +13.3	33/40	0.0321	0.46	6.9 × 10 ⁻⁷
GRAB	shape × fine intent W	17,604	6	40	+1.3	4.5	-0.1 to +2.8	23/40	0.0434	0.30	0.15
GRAB	shape × intent class W	17,604	5	40	+2.3	6.2	+0.3 to +4.2	29/40	0.0436	0.31	0.77
OakInk2	scene × verb W	57,919	5	40	+3.0	7.5	+0.5 to +5.4	28/40	0.0166	0.11	0.78
OakInk2	scene × primitive	57,919	5	40	+2.3	7.3	-0.0 to +4.6	25/40	0.0180	0.12	0.82
OakInk2	annotated transitions † W	57,919	6	12	+16.7	8.4	+11.3 to +22.1	12/12	0.0229	0.12	0.0001

The three OakInk-Image axes with 40 or more seeds and TACO's axis carried a penalty above the control. The two GRAB axes and two of the three OakInk2 axes did not; on those four the 95 % interval on the difference from the control reached at most +3.0 points, which bounds any penalty the design missed there at that size. GRAB's nulls come with a planted control on the same poses (Section 4.1), so under this model, representation, metric and split its held-out shape-by-intent cells showed no penalty. OakInk2 has neither a planted nor a permuted control, and its nulls are reported without interpretation. The category-by-subject axis pairs an object category with the person who grasped it; its penalty shows that the measurement responds to a held-out cell whose second factor is not an action factor, and no claim about composition rests on it.

The four datasets differ in training volume as well as in labelling, so Table 2 is not a controlled comparison between datasets. The two datasets with a penalty have the fewest training windows per arm (4,591 and 8,144, against 17,604 on GRAB and 57,919 on OakInk2), and volume can move the penalty: on OakInk2's scene-by-primitive axis the penalty was +8.56 % at a budget of 64 against +2.30 % at 256 (Appendix B). Volume does not act in one direction, however. On OakInk-Image a budget of 64 gave +10.15 % against +15.19 % at 256, a smaller penalty from less data, and OakInk2's transitions axis carried +16.7 % at the largest volume in the table. GRAB swept at a budget of 64, about 4,400 windows per arm and so matched to OakInk-Image, returned [PENDING: grab_shape_b64, registered in advance, running]. The causal evidence of this study is therefore the within-dataset manipulation of Section 4.3, where volume is held fixed.

Neither the denominator of the penalty nor the informed arm's exposure orders the datasets. The naive arm's error exceeded the informed arm's by 0.0104 on OakInk-Image category by intent and by 0.0035 on TACO, against 0.0006 on GRAB shape by fine intent and 0.0004 on OakInk2 scene by primitive. The informed arm was most exposed on TACO (0.46) and GRAB (0.30 and 0.31), which fall on opposite sides. Holding out a cell also removes part of each of its factors' data from the naive arm: averaged over held cells and both factors, the naive arm lost 31 % to 47 % of a held factor's trajectories on OakInk-Image, 53 % on TACO, 30 % to 31 % on GRAB and 49 % to 52 % on OakInk2, so this loss does not order the datasets either. Within an axis it was weakly related to the per-seed penalty on OakInk-Image ($r = +0.11$ to $+0.37$) and on TACO ($r = +0.25$, $p = 0.12$).

4.3 When the label describes only part of the trajectory

Table 3. OakInk-Image at a budget of 64, 40 seeds each. p is from Welch's test against the zero-truth control at this budget (+2.96 %, 20 seeds).

Version	Penalty (%)	SD	Seeds positive	p
Unmodified clips	+10.15	6.87	36/40	2.1×10^{-7}
Two clips, label describes both	+17.03	11.53	37/40	2.5×10^{-9}
Two clips, label describes the first	+3.87	5.41	31/40	0.37

When the label described only the first clip, the penalty was 13.2 points below the aligned version (95 % CI 9.1 to 17.2, $p = 2.1 \times 10^{-8}$) and 6.3 below the unmodified clips (95 % CI 3.5 to 9.0, $p = 2.2 \times 10^{-5}$), and it was not distinguishable from the control (difference +0.9, 95 % CI -1.1 to 2.9; Table 3). Joining clips did not by itself reduce the penalty: the aligned version was above the unmodified clips (difference +6.9, 95 % CI 2.6 to 11.1, $p = 0.0019$). Without the seeds that failed the coverage gate (2 aligned, 5 misaligned) the two versions read +16.46 % and +4.89 %, still 11.6 points apart ($p = 3.5 \times 10^{-7}$), with the misaligned version against the control at $p = 0.056$.

Misalignment changed both training sets. Averaged over ten seeds, 8.4 % of the naive arm's training clips belonged to held-out cells under another trajectory's label, so the naive arm was contaminated, and 27.9 % of target trajectories contained a clip whose object and intent the naive arm had seen. The share of the informed arm's clips from held-out cells fell from 41.9 % in the aligned version to 30.8 %, so the informed arm was diluted. In the aligned version the naive arm's two figures were 0 %, and 61 % of target clips belonged to a held-out cell against 100 %.

Table 4. Penalty (%) by class of target window, same seeds and settings as Table 3; 40 seeds each. The re-run reproduced the totals of Table 3 (+16.97 % and +3.92 %).

Version	First clip (labelled)	Across the join	Second clip
Label describes both	+17.75 (SD 12.57)	+16.51 (SD 12.70)	+16.07 (SD 11.68)
Label describes the first	+5.64 (SD 6.71)	+5.51 (SD 8.07)	+1.54 (SD 7.58)

In the aligned version the first and second clips did not differ (paired difference +1.7, $p = 0.079$), so position in a joined trajectory did not matter by itself (Table 4, Figure 2). In the misaligned version the first-clip windows, which the label describes and which show a held-out composition in both versions, read 12.1 points below the aligned version's (95 % CI 7.6 to 16.6, $p = 1.3 \times 10^{-6}$), a loss of 82 % of their penalty above the control. They remained +2.7 points above the control (95 % CI 0.3 to 5.0, $p = 0.028$). The two versions consist of different trajectories, so this comparison is between different windows of the

same kind and is not paired. Second-clip windows, which the label does not describe, were a further 4.1 points lower (paired $p = 0.0057$). Most of the loss therefore occurred on windows whose content was of the held-out kind in both versions, which locates it in the training sets rather than in the scoring. The design does not separate the naive arm's contamination from the informed arm's dilution; both follow from a label that describes only part of a trajectory.

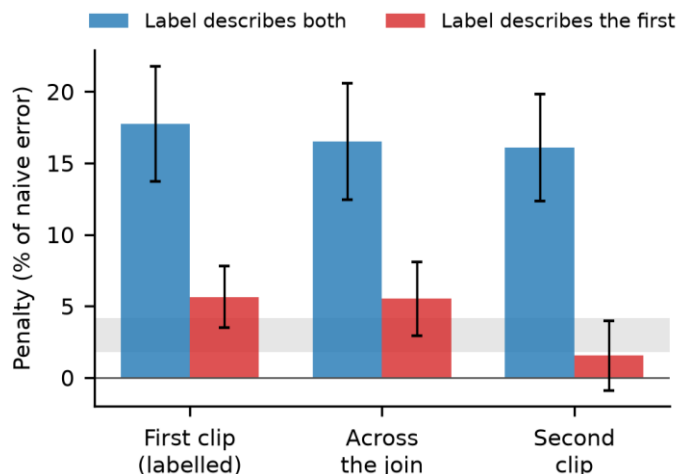


Figure 2. Penalty by class of target window (Table 4), means with 95 % intervals over 40 seeds; the grey band is the 95 % interval of the zero-truth control's mean at a budget of 64.

4.4 The out-of-sample prediction

TACO's trajectories lasted a median of 4.93 s (2.27 to 16.57 s; 1.4 % over 10 s), and its 151 triplets combine 15 actions, 17 tools and 9 objects. As predicted, its action-by-tool axis carried a penalty above the zero-truth control: +10.63 % over 40 seeds, 8.0 points above the control (95 % CI 5.2 to 10.9, $p = 6.9 \times 10^{-7}$), and +10.76 % without the 6 seeds that failed the coverage gate. It was also 14.0 points above TACO's permuted grid (Table 1).

The per-seed estimates were widely spread. They ranged from -3.76 % to +27.43 %, with an interquartile range of +4.32 % to +15.77 %; 8 of 40 fell at or below the mean of the zero-truth control, and 7 exceeded +20 % (Figure 1). A seed changes the held-out cells, the random fill of both arms and the training noise together, and these components were not separated. The spread is therefore that of this paired estimate, not of the split alone. The zero-truth control's per-seed SD of 2.30, which contains the fill and training components on synthetic data, suggests that most of TACO's SD of 8.33 comes from which cells are held out, but that was not tested on TACO itself.

5. Discussion

This study asked when a held-out label combination measures compositional generalisation on hand-motion data. On OakInk-Image, making a correct label describe only half of its trajectory, with the number of trajectories, the categories and the training volume held fixed, removed a penalty of +17.0 % to a level not distinguishable from the control. The loss occurred mostly on windows that showed a held-out composition in both versions, so it lay in what the training sets contained. A penalty on TACO, whose labels each name one tool-use action, was predicted before its data were seen and was found, above both

the synthetic control and TACO's own permuted grid. The mechanism is that the split is made on labels: when a label describes only part of a trajectory, the rest of the trajectory can carry a held-out composition into the naive arm under another label, and the informed arm's held-out trajectories are partly other motion.

OakInk2's transitions axis is consistent with this reading and was not designed to test it. There a recording's label lists every primitive it contains, so a recording that contains a held-out transition is excluded from the naive arm wherever the transition occurs. The naive arm is then free of the held-out motion, although most of each scored target recording is other motion, and the axis carried the largest penalty in Table 2 (+16.7 %, 12 seeds). Contamination of the training sets, not the purity of what is scored, is what Table 4 and this axis both point to.

Three points follow. First, a label that describes the scored motion was necessary for the penalty on OakInk-Image, but it is not sufficient for one: GRAB trajectories trimmed to their contact segment still returned a penalty at the control (+2.22 %, $p = 0.78$; Appendix B). Where a dataset annotates the frame span of each action, contamination can be measured directly. On OakInk2's scene-by-primitive axis the labelled primitive covered a median of 57 % of a recording's frames, 259 of 609 recordings held a single primitive, and over the sweep's 40 splits 11.6 % of the naive arm's training recordings contained a held-out scene and primitive after their first primitive (4.9 % of its training frames), while 40 % of target frames were the held-out primitive. Second, a paired penalty has a floor that differs between datasets. The design returned +2.6 % when the true value was zero in synthetic data, and a permuted grid supplied a floor on each real dataset where one could be built; a test against zero would have declared GRAB's shape-by-intent-class axis (+2.3 %, $p = 0.77$ against the control) a compositional penalty. Third, the estimate must be read over many seeds. On TACO a single seed could have returned any value from -3.8 % to +27.4 %, so a comparison between models or splits that rests on one partition and one run cannot be read as a property of either.

These points give a short protocol for a held-out-combination split of observational motion. Check that each label describes the motion in the scored windows, and cut recordings at annotated boundaries or report contamination where it can be measured. Hold out whole fine labels and check label, frame and constituent leaks. Report the penalty against a zero-truth control and a permuted grid on the same dataset rather than against zero. Recover a planted effect before interpreting a null. Report training volume and the distribution over many seeds.

Several limitations apply. The error is an autoencoding error, so the results show that a prior passes an unseen composition through its bottleneck less well when it has not seen that cell, not that it cannot generate one. One prior architecture was used, so the size of the penalty, though not necessarily its pattern, may differ for others. The condition was manipulated on one dataset and predicted on one other, and differences between datasets are confounded with training volume. The naive arm's contamination and the informed arm's dilution were not separated, and in the misaligned trajectories the label was also less informative about intent over the whole trajectory. A permuted cell is not as compact in pose space as a real one, so an unseen composition was not separated from an unseen region of pose space. The aligned version exceeded the unmodified clips, possibly because a budget counted in trajectories gives the informed arm twice as many held-out clips when each trajectory holds two; this was not tested.

In conclusion, a hand-motion dataset can test compositional generalisation only when its labels describe the scored motion. When a correct label described only part of a trajectory, holding out the label left the

motion in the training sets and the penalty disappeared; on TACO, whose labels each describe one action, a penalty registered in advance was found. Whether a model appears to generalise to unseen compositions therefore depends on how the data were labelled and on how the measurement was calibrated, as well as on the model.

Broader impact statement

This work concerns the evaluation of motion models and has no direct application. Its main risk is the one it addresses: a compositional score read without calibration can overstate or understate what a model has learned.

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Appendix A. Datasets and conventions

Flexion axis, flexion sign and abduction axis were inferred per dataset from joint-limit satisfaction rather than taken from documentation. On TACO this inference contradicted the dataset's own loader: without adding MANO's mean pose, 5.84 % of values fell outside a joint limit over all 2,317 sequences, and with it 15.99 %, so the mean pose was not added. Degrees of freedom pinned at a limit or never moving numbered 1 of 27 on TACO and between 1 and 6 on the other datasets; the error is computed over all 27. A held-out cell had to span at least four distinct fine labels (five on GRAB's shape-by-fine-intent axis), and 12 to 60 cells per axis were eligible, so the spread over seeds understates the uncertainty over which cells were held. Coverage failed on 16 of 130 seeds of affordance by intent, 5 of 69 of category by intent, 3 of 12 of functional class by intent, 6 of 40 on TACO and at most 3 of 40 elsewhere.

Appendix B. Further experiments

Retargeting quality and grid density. OakInk-Image poses were clamped so that six degrees of freedom were pinned or dead, as in GRAB, and whole cells of its category-by-intent grid were removed at random to reduce occupancy from 76 % to 49 % (which also removed 8 categories and 243 of 770 trajectories). The penalty was +13.70 % and +19.64 % respectively, against +15.19 % unmodified ($p = 0.39$ and $p = 0.0025$).

Stride and trimming on GRAB. At a stride of 32 the unmodified GRAB axis did not differ from its stride-4 value in Table 2 ($p = 0.71$). Keeping only the contact segment of each GRAB trajectory (8.5 s to 4.9 s; 58 of 1,048 trajectories dropped; about 11,400 windows per arm) gave +2.22 %, not different from the control ($p = 0.78$) or from the unmodified sweep ($p = 0.43$).

Budget on OakInk2. Whole OakInk2 recordings on the scene-by-primitive axis returned +2.30 % at a budget of 256 and +8.56 % at a budget of 64 (difference +6.3, $p = 0.0097$; the latter above its control, $p = 0.011$). Recordings cut at their annotated primitive boundaries and labelled by their own primitive returned +8.93 % at a budget of 256, which matches the frame volume of whole recordings at 64 and did not differ from them ($p = 0.89$), and +5.19 % at a budget of 896, which matches the frame volume of whole recordings at 256 and was not above them (difference +2.9, 95 % CI -0.5 to 6.3, $p = 0.097$). On OakInk2, therefore, cutting recordings changed the penalty no more than the volume of training data did, and the whole-recording null is reported without a cause.

Appendix C. Departures from plan

The affordance-by-intent sweep was extended from 70 to 130 seeds after its first 70 had been seen; this is the one exception to the rule of Section 3.8, and no claim rests on that axis alone. An interim value of the category-by-subject sweep was seen at 28 of 40 seeds through a scripting side effect; its seed list and readings were not changed. Window stride was not uniform in the earliest sweeps; the two affected axes were re-run at the common stride, which left the GRAB axis unchanged and turned the OakInk2

transitions axis from +3.44 % at a stride of 16 into the penalty of Table 2. A first version of the joined OakInk-Image trajectories reused source clips, so that target frames appeared in training; it was discarded and the frame-leak gate was added. A second version joined clips of different objects under one fine label, which exposed the informed arm to the targets' objects; it was replaced by the version of Table 3 and the constituent-leak gate was added. A test of whether cutting OakInk2 recordings at primitive boundaries creates a penalty was withdrawn when its declared data-volume control reproduced the effect (Appendix B). The two † rows of Table 2 were run at 12 seeds in an earlier phase; their replications at 40 seeds were declared before they started.

Appendix D. Every sweep run under the protocol

Sweeps on real data not reported in the main text, with their results (the prior of Section 3.2; penalty as a percentage of the naive error; budget 256 and stride 4 unless stated). Sweeps of a second prior architecture, run for a comparison outside the scope of this paper, and development runs on synthetic data are listed in the supplement.

Sweep	n	Penalty (%)	Why not in the main text
OakInk-Image category $\times$ intent, second set of models on the same 70 seeds	70	+14.78	Trained for a comparison with a second architecture; agrees with Table 2
OakInk-Image object-id prefix $\times$ intent, four sweeps	18 to 70	+14.30 to +17.81	The prefix marks a sub-collection, not a category; superseded by the official taxonomy
OakInk-Image object $\times$ intent, budgets 256 and 64	5	+5.56 and -2.22	2.2 trajectories per cell, too few to split
DexYCB subject $\times$ object, two versions	12	+0.09 and +1.12	No action factor; early split code
GRAB shape $\times$ fine intent with contact channels, two inputs, stride 32	20	-0.03 and +0.90	Different input channels
GRAB shape $\times$ fine intent, first run, stride 32	2	-10.12	Stopped; replaced
GRAB contact segment, stride 4	7	-1.35	Stopped to free the GPU; the stride-32 sweep is in Appendix B
OakInk2 annotated transitions, stride 16, two versions	12	+9.06 and +3.44	Stride 16 and early split code; replaced at stride 4
OakInk2 primitive segments, two further sweeps	20 and 40	+8.97 and +7.14	Confounded with training volume (Appendix B)
OakInk-Image joined clips, five earlier versions, budgets 256 and 64	11 to 40	+3.03 to +57.24	Reused clips or leaked constituents (Appendix C)
OakInk2 subject $\times$ verb, three held cells	-	-	Started; not completed

Declaration of AI assistance

A large language model was used beyond language editing: it implemented and ran the analysis scripts, launched the training runs and drafted text under author direction. The authors set the research questions, decided which claims to make and are responsible for the content. Every reported number is produced by a script operating on stored per-seed results, both supplied, so any figure can be recomputed independently of the tool.